

F450 Payload Layout

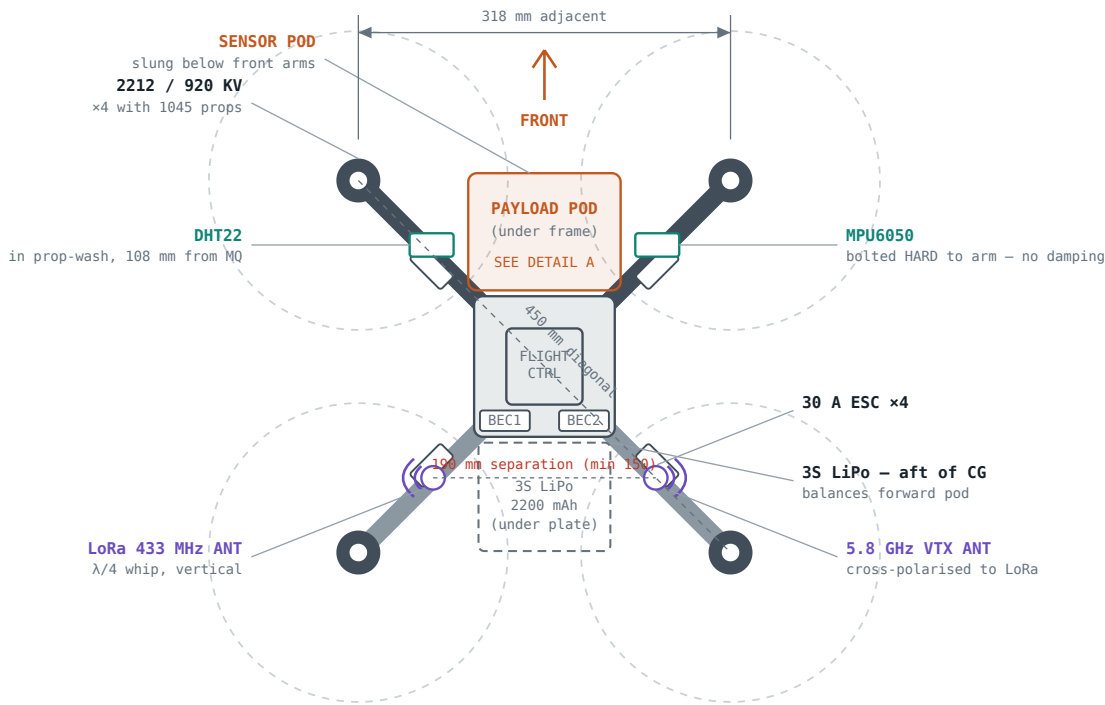
Physical arrangement of the sensor payload on the airframe — where every component mounts, which way it points, and why that position and no other.

⚠ This diagram was drafted with AI assistance (Claude) from our own component list, mass budget and mounting constraints — it is a planning sketch, not a photograph or CAD export of the built hardware. Sensor placements shown here follow the mounting schedule below and are being verified against the physical build as it's assembled.

AIRFRAME F450 · 450 mm PAYLOAD MASS 155 g AUV ≈1.30–1.45 kg T/W 2.07 : 1
SCALE 0.8 px : 1 mm REV 1.0

VIEW 1 Plan view — component placement

X-configuration · viewed from above · dashed outlines sit below the frame



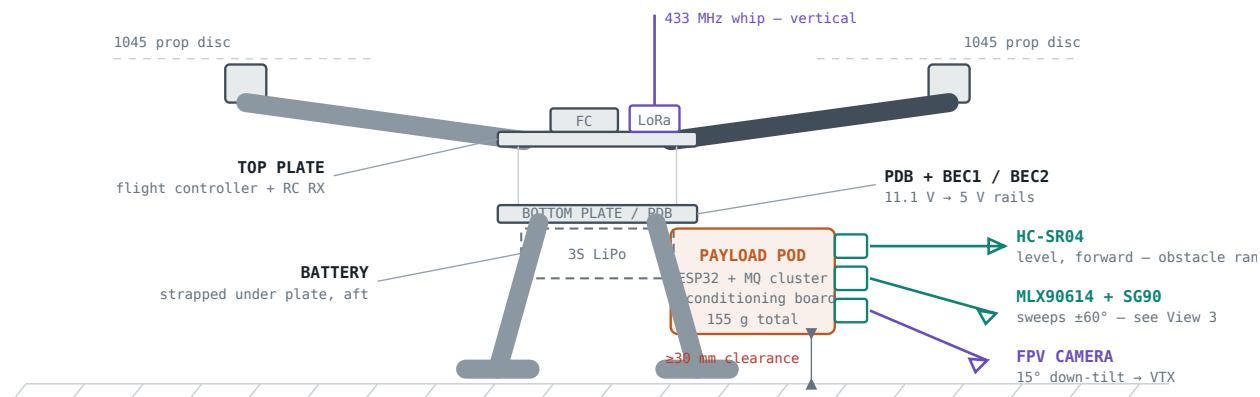
VIEW 1 — PLAN

All dashed outlines are components mounted beneath the frame plates.
Forward pod + aft battery place the CG within 10 mm of the frame centre.

— airframe / flight domain — payload pod & 5 V — sensors on 3.3 V — RF / antennas
— power source

VIEW 2 Side elevation — stack height and sensor aim

nose to the right · arrows show each sensor's line of sight

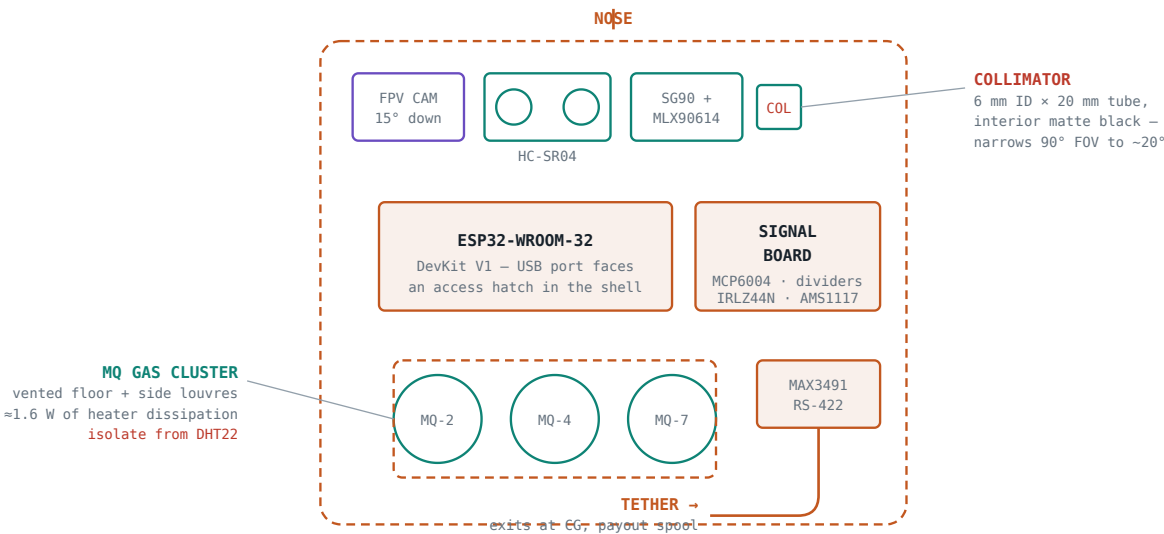


VIEW 2 – ELEVATION

Pod hangs forward of the battery so the two masses straddle the CG.
Landing gear must clear the pod – extend the stock legs or the pod grounds out on touchdown.

DETAIL A Payload pod — internal arrangement

viewed from below, nose upward · enlarged 1.8×



DETAIL A – POD INTERIOR

Heat-producing parts (MQ cluster, AMS1117) sit aft and low; the analog signal board sits away from the servo's current path.

VIEW 3 Thermal scan geometry

why the collimator earns its place

BARE MLX90614 – 90° FOV



footprint at 3 m
28.3 m² – target fills 1.8 %
rise = +0.3 °C
below sensor noise → miss

SERVO SWEEP

5–7 stops across ±60°,
300 ms settle before each read.

Each stop is one thermal sector,
not one pixel – describe the output
as a coarse sector scan.

WITH 6 mm × 20 mm COLLIMATOR – 20° FOV



footprint at 3 m
2.03 m² – target fills 24.6 %
rise = +3.6 °C
12× the signal → confident detect

Mounting schedule

Position is a design decision, not a convenience — each row states the constraint that fixes it.

COMPONENT	MOUNTS TO	ORIENTATION	WHY THERE
● MPU6050	front-right arm	flat, axes aligned	Rigidly bolted — no foam, no gel. This is not the flight IMU; frame vibration is the signal it exists to measure. Damping it would erase the data.
● DHT22	front-left arm	vents downward	108 mm from the MQ cluster and inside prop-wash. Any closer and the ~1.6 W of heater dissipation biases it by several degrees.
● MQ-2 / MQ-4 / MQ-7	pod floor, aft bay	sensing face down	Needs airflow, so the bay is vented on the floor and both sides. Placed aft and low so

COMPONENT	MOUNTS TO	ORIENTATION	WHY THERE
			its heat rises away from the analog board.
● MLX90614 + SG90	pod nose, right	sweeps $\pm 60^\circ$ horizontal	Needs unobstructed line of sight past the pod shell and clear of the MQ bodies, which are warm and would sit inside the cone.
● HC-SR04	pod nose, centre	level, dead ahead	Centred so its reading corresponds to the flight path. Level rather than tilted, or it ranges the floor instead of obstacles.
● FPV camera	pod nose, left	15° down-tilt	Standard FPV tilt for forward flight. Electrically isolated from the ESP32 — this is an independent analog chain.
● LoRa Ra-02	top plate, aft-left	whip vertical	Antenna vertical for an omnidirectional pattern in the tunnel plane, and 190 mm from the VTX antenna.
● 5.8 GHz VTX	top plate, aft-right	antenna up, cross-pol	Maximum separation from the LoRa whip. The risk

COMPONENT	MOUNTS TO	ORIENTATION	WHY THERE
			is the VTX regulator's broadband noise, not the band itself.
● ESP32 + signal board	pod, mid bay	USB toward hatch	Reflashing without disassembling the pod. Analog board kept away from the servo's high-current return path.
● BEC1 / BEC2	bottom plate	beside PDB	Short, fat leads to the battery tap. Two separate BECs so the VTX's noise never reaches the sensor rail.
● 3S LiPo	under bottom plate	aft of centre	Counterweights the forward pod. Slide it fore/aft on the strap to trim the CG after the pod is loaded.
● Tether anchor	pod belly, at CG	trails aft	Anchored at the CG so tether tension produces no pitch moment. Anywhere else and the drone fights the cable.